

Does Your EDR Do That?

Or: Not All EDR Products Have All The Features You'd Expect



- Chris Taylor
 - Founder and lead security consultant at Taksati Consulting
 - 25+ years of experience in digital forensics, incident response, threat intelligence, and so on
 - Builds and runs security programs for small media and entertainment companies
- Caveat emptor
 - You all know way more about Linux than I do
 - I will tell you the requirements and leave to you how best to meet them
- Any mention of a product by name is **NOT** an endorsement of that product, just illustrative examples

A Short Walk Through History...

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- Anti-Virus (AV)
- Anti-Malware (AM)
- Next Gen Anti-Virus (NGAV)
- Endpoint Detection and Response (EDR)
- Extended Detection and Response (XDR)

A Short Walk Through History...

- Anti-Virus (AV)
 - First Virus = Creeper infected a bunch of PDP-10s in 1971 (technically a worm, not a virus)
 - First Anti-Virus = Reaper by Ray Tomlinson, written specifically to remove Creeper
 - First use of the term “computer virus” = 1983 by Fred Cohen
 - First commercial AV product = Sophos in 1985
 - First commercial AV in USA = McAfee in 1987 (but that’s a different story)
 - First heuristic AV product = Flushot Plus or Anti4us in 1987, F-PROT in 1991 is first like modern
 - First virus on Linux = Staog in 1996
- Anti-Malware (AM)
- Next Gen Anti-Virus (NGAV)
- Endpoint Detection and Response (EDR)
- Extended Detection and Response (XDR)

A Short Walk Through History...

- Anti-Virus (AV)
- Anti-Malware (AM)
 - Virus = self replicating, appends to a file or system area
 - Worm = self replicating, travels via network usually using some exploit
 - Trojan = a stand-alone malicious file pretending to be safe, legit file
 - Rootkit = a collection of trojans and malicious kernel mods used to maintain access and hide
 - Every other type of malware is a Trojan defined by function rather than by type, e.g. 'backdoor'
 - Trojans are way more common than viruses, thus 'anti-virus' became 'anti-malware'
 - It now does a lot more than just AV (HIDS, DLP, etc), thus 'anti-malware' became 'endpoint security'
- Next Gen Anti-Virus (NGAV)
- Endpoint Detection and Response (EDR)
- Extended Detection and Response (XDR)

A Short Walk Through History...

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TECHNICAL SECURITY SERVICES

- Anti-Virus (AV)
- Anti-Malware (AM)
- Next Gen Anti-Virus (NGAV)
 - ~100,000,000 new malwares released every year
 - Signature databases got huge, always lagged, generally struggled to keep up
 - AV companies started leaning harder on the heuristics and behavior-based detections
 - First NGAV = Cylance released in 2012
 - There is no single standard definition of what “next gen” means
 - NGAV still relies on signatures, but heuristics engine is mo’ stronger
- Endpoint Detection and Response (EDR)
- Extended Detection and Response (XDR)

A Short Walk Through History...

TAKSATI CONSULTING
TECHNICAL SECURITY SERVICES

- Anti-Virus (AV)
- Anti-Malware (AM)
- Next Gen Anti-Virus (NGAV)
- Endpoint Detection and Response (EDR)
 - First EDR = Carbon Black released in 2011
 - First use of term “EDR” = by Anton Chuvakin at Gartner in 2013
 - Originally designed to complement AV; it did NOT block malware, it enhanced response capabilities
 - Originally designed to compete with enterprise forensics, but be way faster
 - Collects ‘telemetry’ = logs every process start, network session, file access, etc.
 - Customers wanted less agents and less invoices... so NGAV and EDR merged
- Extended Detection and Response (XDR)

A Short Walk Through History...

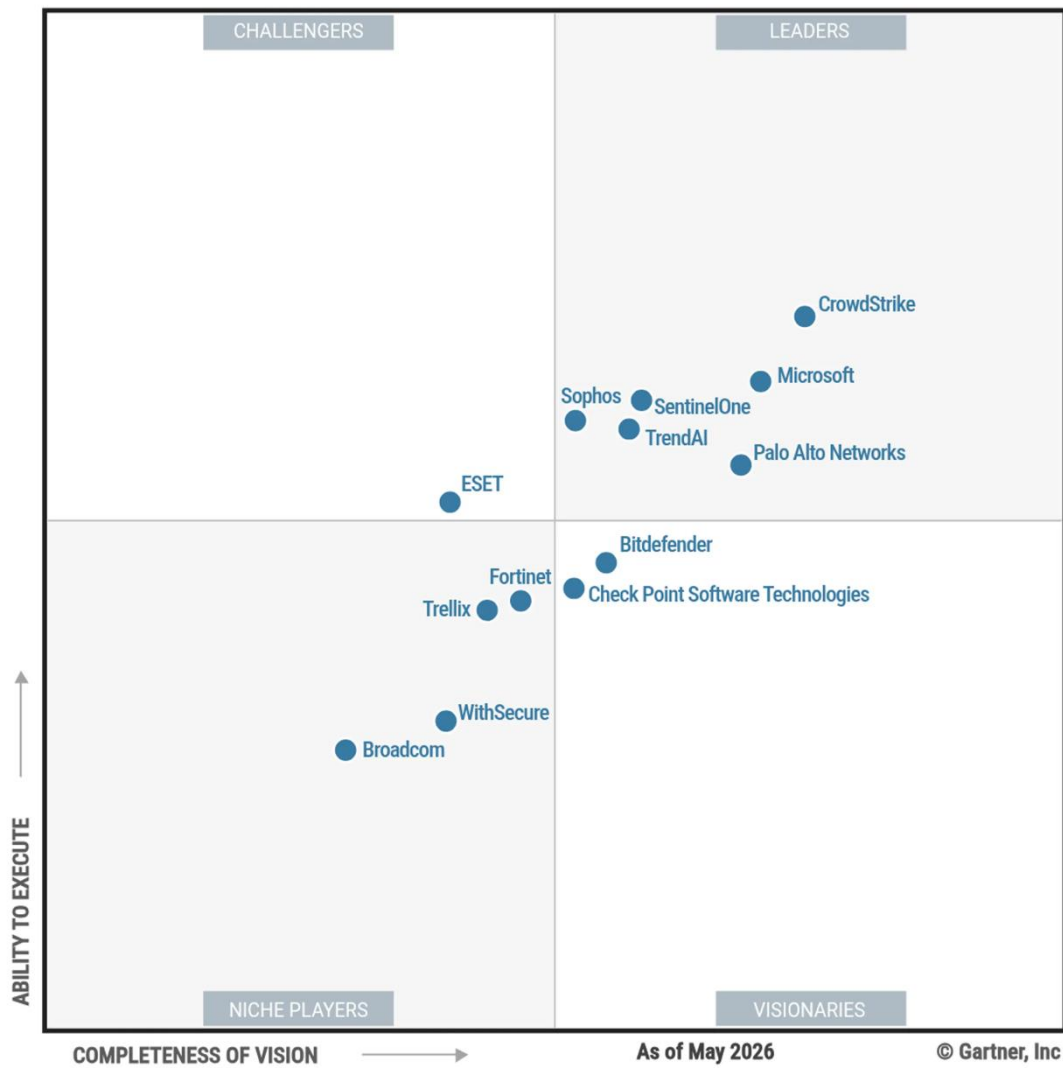
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- Anti-Virus (AV)
- Anti-Malware (AM)
- Next Gen Anti-Virus (NGAV)
- Endpoint Detection and Response (EDR)
- Extended Detection and Response (XDR)
 - First use of “XDR” = by Nir Zuk of Palo Alto Networks in 2018
 - Full enterprise visibility requires monitoring logs, network, and endpoint
 - Those used to be three isolated silos
 - Combining them greatly enhances visibility and correlation of events
 - Tools that combine them call themselves XDR

EDR Market

THE FORRESTER WAVE™

Extended Detection And Response Platforms
Q2 2026



Which should I choose?

- It depends...
 - What are your needs?
 - What are your desires?
 - What is your budget?
 - What is your environment?
 - ...
- Choosing a product in four easy steps:
 - Step 1: develop measurable requirements
 - Step 2: compare options against those requirements, actually test them
 - Step 3: select the product that best meets those requirements
 - Step 4: figure out how to pay for it (that's a different presentation)

EDR Telemetry

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- <https://www.edr-telemetry.com/>
- A project by Defendpoint Consulting – I am not affiliated with this project in any way
- They tested a variety of EDR based on a standard grading criteria
 - Provides an apples-to-apples comparison
 - What telemetry do they collect?
 - What features do they have?

Telemetry Scores

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Rank	Windows		Mac		Linux	
	Product	Score	Product	Score	Product	Score
1	C-Prot	39.60	Phorion	35.00	C-Prot	25.60
2	Harfanglab	37.35	C-Prot	34.40	Uptycs	23.80
3	Uptycs	36.60	Elastic	19.80	CrowdStrike	16.95
4	Cortex XDR	35.80	CrowdStrike	14.60	Elastic	15.80
5	SentinelOne	35.75	MDE	13.70	SentinelOne	15.60
6	CrowdStrike	35.45	LimaCharlie	13.60	BitDefender	14.60
7	MDE	34.80	ESET Inspect	13.50	MDE	13.70
8	Elastic	33.45	BitDefender	12.90	LimaCharlie	12.70
9	ESET Inspect	30.80	Uptycs	12.30	Auditd	12.00
10	Trellix	29.60	Qualys	8.40	Cortex XDR	11.4

Notice how the scores differ by platform – the best Linux EDR scores lower than the 10th best Windows EDR

Telemetry Categories

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Windows

Process Execution
File Operations
Hash Algorithms
Network Connections
User Account Activity

Scheduled Task Activity
Service Activity
EDR SysOps
PowerShell Activity
Driver/Module Activity

Device Operations
Registry Changes
Named Pipe Activity
WMI Activity
BITS Jobs Activity
Miscellaneous System Changes

Mac

Process Activity
File Activity
File Metadata
Network Activity
User Account Activity
User & Session Activity
Scheduled Task & Persistence Activity
Service Activity
EDR SysOps
Script Activity
System Extension & Driver Activity
Process Tampering Activity
Access Activity
Device Activity

Code Signing & Trust Activity
Privacy & TCC Activity

Linux

Process Activity
File Manipulation
Hash Algorithms
Network Activity
User Account Activity
User Activity
Scheduled Task Activity
Service Activity
EDR SysOps
Script Activity
Driver/Module Activity
Process Tampering Activity
Access Activity

Telemetry Comparison

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		Linux			Windows		
		auditd	Carbon Black	CrowdStrike	sysmon	Carbon Black	CrowdStrike
Process Acty	Created	Y	Y	Y	Y	Y	Y
	Terminated	N	N	Y	Y	P	Y
	Tampered	Y	N	N	Y	P	Y
	Accessed				Y	Y	Y
	Loaded				Y	Y	Y
	Remote				Y	Y	Y
	Call Stack				Y	N	Y
	W32 API				N	Y	Y
File Access	Created	Y	Y	Y	Y	Y	Y
	Modified	Y	Y	Y	N	Y	Y
	Deleted	Y	Y	N	Y	Y	Y
	Opened				N	Y	Y
	Renamed				N	Y	Y
Network	Connection	Y	Y	Y	Y	P	Y
	Listening	Y	N	Y			
	DNS	N	N	Y	Y	Y	Y
	File Download				N	N	Y
	URL				N	N	N
Service	Created	N	N	P	N	P	Y
	Modified	N	N	P	N	N	P
	Deleted	N	N	N	N	N	N

auditd

- <https://github.com/linux-audit/audit-userspace>
- <https://man7.org/linux/man-pages/man8/auditd.8.html>
- Goal is to provide tamper-resistant audit logs that support security monitoring and incident investigation to make Linux compliant with Common Criteria, PCI-DSS, etc.
- Logs system calls, file access, authentication events, and configuration changes
- Uses configurable audit rules to monitor specific files, directories, users, processes, etc.
- Commercial EDR tools either
 - use it for a portion of their telemetry gathering or
 - require either tuning it or disabling it when installing their tool to prevent them from fighting

dnotify -> inotify -> fanotify

- Linux Kernel subsystem that provides monitoring of file system activity
- dnotify
 - monitors directory-level changes for file creation, modification, deleted, or renamed events
- inotify
 - monitors both files and directories and covers a broader range of events than dnotify
 - is optimized for tracking changes to specific files or directories, making it useful for file synchronization, configuration monitoring, and application automation.
- fanotify
 - provides system-wide file access monitoring and can intercept file open and access events.
 - is designed for security and antivirus use cases, allowing applications to inspect and optionally block file access before it occurs.
- Compared to inotify, fanotify provides broader visibility across filesystems and greater control over file access decisions
- Many commercial EDR and AV products use this for file access monitoring and blocking

Netfilter and nftables

- Linux Kernel subsystem that provides monitoring of network activity
- Netfilter is the engine within the kernel // nftables is the interface to configure that engine
- iptables is the legacy interface that has been replaced by the more modern and efficient nftables
- Provides a host-based firewall and IDS that can monitor and block network activity
 - Doesn't provide the IDS... it provides the IDS with efficient access to the traffic, something else provides detection rules
- Many commercial EDR products use this for network monitoring and blocking

- <https://ebpf.io>
- Extended Berkeley Packet Filter, a Linux Foundation Project, first extended in 2014
- Wraps programs to monitor system and kernel events to provide observability, networking, performance monitoring, and security
- Provides low-overhead, real-time visibility into processes, system calls, file activity, and network connections without requiring traditional kernel modules.
- Offers improved performance, flexibility, and compatibility compared to legacy monitoring mechanisms such as auditd, ptrace, and loadable kernel modules
- Several commercial EDR tools use this as a primary source for their telemetry

osquery

- <https://osquery.io>
- Developed by Facebook, first released in 2014
- Goal was to enhance endpoint monitoring and make enterprise-wide searches easier
- Exposes low-level OS system data as if it was a SQL database
 - Uses SQL language to return data from endpoints
 - `SELECT process.name, process.pid, cmdline FROM processes WHERE path LIKE 'C:\Users\%';`
- Facebook and Linux Foundation created a new foundation to host and control its code base
 - Open sourced all the code
 - Plugin framework allows it to be customized and extended
- Several commercial EDR tools are a custom frontend on osquery or use it for searching

- <https://github.com/microsoft/SysmonForLinux>
- Developed by Microsoft, ported to Linux, released as FOSS
- Generates structured event logs for activities such as process creation, network connections, file modifications, and module loading.
- Uses customizable XML-based configuration rules to selectively capture high-value security events.
- Integrates with SIEM and security monitoring platforms to support threat detection, incident response, and forensic investigations.
- Provides telemetry similar to Windows Sysmon, enabling consistent monitoring across Windows and Linux environments.

In summary...

When it comes to EDR features:

- Selecting a tool isn't just which is best
 - you want the one that best for your specific requirements
- The feature disparity between Windows and Linux is huge, and between tools is notable
 - Make sure the tool you select has the features you need
- Tools only have to partially support one feature to claim support for a whole category
 - Don't assume, test
- It's good to know what the “under the hood” components are your tools are depending on
 - Supply chain risk management, software bill of materials

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Chris Taylor
Taksati Consulting
info@taksati.org